

Department of Electronic & Telecommunication Engineering

University of Moratuwa, Sri Lanka

EN3160 - Assignment 02

Fitting and Alignment in Computer Vision

Submitted by

230145E Dilhan W.A.

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EN3160

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1. Sunflower detection using scale-space extrema

Repository: `https://github.com/ahesh-dilhan/en3160-assignment-02`

The supplied image is read with `cv.IMREAD_REDUCED_COLOR_4`, giving 360×360 pixels. Dark central disks are detected using darkness $D = 1 - I_{\text{gray}}/255$ and scale-normalized LoG response $L = -\sigma^2 \nabla^2 (G_\sigma * D)$. Positive spatial/scale maxima indicate dark blobs; their ideal-disk radius estimate is $r = \sqrt{2}\sigma$.

```
L = np.stack([-s*s*gaussian_laplace(D, s) for s in sigmas])
peaks = (L == maximum_filter(L, size=(3, 3, 3))) & (L >= 0.12)
```

Parameters. Use 36 geometric scales, $\sigma \in [1.5, 16]$ pixels, excluding end-scale maxima; threshold 0.12; permitted overlap 35% of the smaller disk area. Yellow-petal support must cover 50% of the annulus $[r, 2.2r]$ with angular coverage. A Hessian curvature ratio ≥ 0.5 rejects elongated gaps. No Hough-circle detector is used.

(a) Reduced input: 360 x 360



(b) LoG detections; largest five numbered



Rank	x	y	σ	r	Response
1	282	338	7.106	10.050	0.351
2	106	255	5.801	8.204	0.365
3	179	327	5.801	8.204	0.346
4	178	261	5.422	7.668	0.353
5	274	258	4.736	6.698	0.347

Coordinates and radii are in reduced-image pixels.

Figure 1: Input and 53 candidates. The largest five disks are numbered.

Interpretation. The largest disk is at $(282, 338)$, $r = 10.05$ pixels. The origin is top-left; multiply coordinates and radii by four for original-image units. The prior suppresses foliage but misses distant, tilted or occluded flowers. These are candidates, not a verified count; radii describe central disks rather than complete blossoms.

2. Custom RANSAC: line followed by circle

Data. Listing 1 is reproduced with seed 0: 50 circle points with centre $(2, 3)$, radius 10 and radial-noise standard deviation 0.625, plus 50 points from $y = -x + 2$ with vertical-noise standard deviation 1. Both loops are hand-written: two distinct points define a line hypothesis; three non-collinear points define a circle. Degenerate samples are rejected. Each model uses 2,000 trials; the sequence starts with RNG seed 100.

Residuals and consensus. Represent the line by $ax + by = d$ with $a^2 + b^2 = 1$, $d \geq 0$. The errors are

$$e_\ell(\mathbf{p}) = |ap_x + bp_y - d|, \qquad e_c(\mathbf{p}) = \big|\|\mathbf{p} - \mathbf{c}\|_2 - r\big|.$$

Both thresholds are 1.5: approximately 2.12 normal-noise standard deviations for the true line and 2.4 for the circle. Require at least 35 line inliers, remove that consensus, and require at least 25 circle inliers from the remainder. TLS refines the line; L-BFGS-B minimizes mean squared radial error with $r > 0$. Membership is updated after refinement; winning samples and unrefined models are retained separately.

```
line = ransac(X, "line", rng, threshold=1.5, minimum_consensus=35)
remaining = np.flatnonzero(~line.inliers)
circle = ransac(X[remaining], "circle", rng,
                threshold=1.5, minimum_consensus=25)
```

For $p = 0.99$, $w = 0.5$ and sample size s, $N = \lceil \log(1-p)/\log(1-w^s) \rceil$ gives 17 line or 35 circle trials. The larger fixed budget accommodates contamination and reduced inlier fractions; it does not guarantee success.

Model	Refined parameters	Inliers	Inlier RMSE
Line	$a = 0.7210,\ b = 0.6930,\ d = 1.5724$	54	0.6706
Circle	$c_x = 1.9494,\ c_y = 2.9879,\ r = 9.8919$	46	0.5925

Circle first. Seed 0 also succeeds: centre $(1.943, 3.008)$, radius 9.930. For data seeds 0–19, circle recovery occurs in 20/20 line-first and 19/20 circle-first runs, defining recovery as centre and radius errors below 2. A large-radius circle can approximate a line and steal its consensus, although circle-first need not fail. Line-first can remove genuine circle points near intersections, explaining its 54-point consensus. Refinement is local, not a proof of global optimality.

RANSAC results and fitting-order comparison

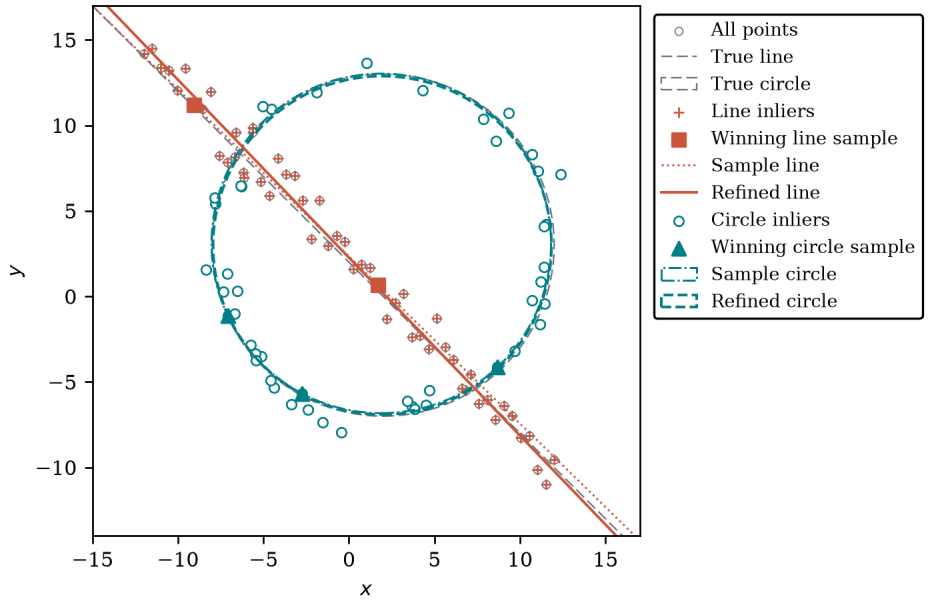


Figure 2: All points, winning samples, sample hypotheses, refined models and their line/circle consensus sets, shown together.

3. Planar homography and image compositing

Two architectural photographs are paired with original graphics: an open-studio banner on a flat facade band (oblique view) and a community-garden notice on a garage door (nearly frontal). Both notices are fictional. Four destination corners are ordered TL, TR, BR, BL. They were selected visually, not recorded mouse clicks; `select_corners.py` provides four-click selection.



Figure 3: Composited results for the two chosen image pairs, retaining original colours. Inputs and original overlay graphics are available in the repository.

The four source/destination pairs define $\mathbf{x}' \sim H\mathbf{x}$, mapping overlay to background. Warp the graphic and an all-one coverage mask with the same bilinear interpolation. With premultiplied warped colour W and coverage M, $C = B(1 - \alpha M) + \alpha W$; black graphic pixels remain valid content.

```
H = cv.getPerspectiveTransform(source_corners, destination_corners)
result = background*(1-alpha*M[..., None]) + alpha*W
```

Pair	Destination corners (pixels)	α
Studio	(215,405), (953,508), (953,556), (215,461)	0.82
Garden	(454,683), (782,683), (786,800), (450,800)	0.78

Corner reprojection error is below 0.001 pixels, confirming correspondence rather than independent accuracy: the same points define H. Alpha blending retains texture but does not model physical illumination, shadows, reflections or occlusion. The facade placement is more distorted than the garage placement.

Image credits. Expect Best, Pexels 323780; Binyamin Mellish, Pexels 1396122. Downloaded at width 1600; original graphics added. Pexels license; full attribution in the repository.

4. SIFT-based PCB registration and differences

Registration. The template is `AECC1.jpg`. SIFT extracts up to 8,000 features per image at maximum side 1600. `BFMatcher` uses L2 two-neighbour matching with a 0.72 ratio test. RANSAC estimates the target-to-template homography with a 3-pixel threshold at feature scale, confidence 0.999 and up to 10,000 trials. The mapping is converted to original-image coordinates before warping all five targets.

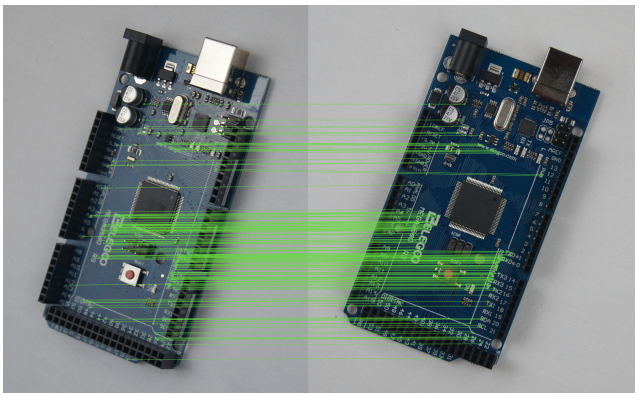


Figure 4: ADAD1 to AECC1: green lines show accepted RANSAC matches.

Registration confidence. At least eight geometric inliers are required. AAAA1 has only 15, with an inlier hull spanning 10.1% of the full template image, so its large viewpoint change makes it less reliable than AECC2–4. Full-size match visualizations and alignment overlays are saved in the experiment output folders.

Valid overlap. An all-one image is warped with the same interpolation. Partly padded pixels and a one-pixel boundary margin are excluded. This prevents black warp borders from inflating the difference statistics.

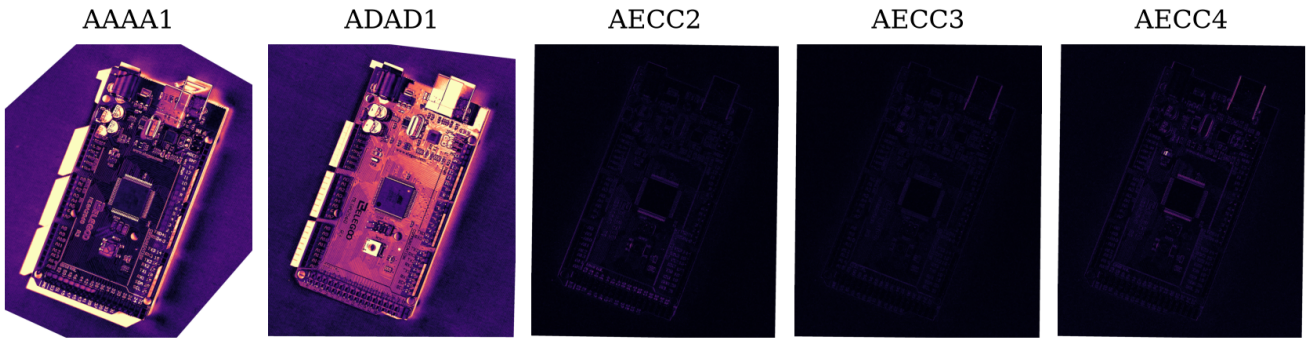


Figure 5: Mean absolute RGB differences, displayed with the original magma colour map over a common 0–100 range: dark is zero and pale yellow is 100 or above. Blank white margins outside the warped footprint are invalid overlap. Numerical metrics use the full raw values.

Target	Matches	Inliers	Median error*	Overlap	Raw MAE
AAAA1	38	15	0.944	89.0%	29.31
ADAD1	252	187	0.439	98.1%	30.62
AECC2	3314	3068	0.210	98.3%	3.52
AECC3	3978	3664	0.167	98.8%	3.29
AECC4	3171	2913	0.231	98.4%	4.07

*Median inlier reprojection error, in feature-image pixels. MAE averages valid RGB-channel differences over the entire overlap, including background. It is not a board-only defect score. No exposure normalization is applied.

```
H_full = np.linalg.inv(template_scale) @ H_small @ source_scale
aligned = cv.warpPerspective(target, H_full, template_size)
difference = cv.absdiff(aligned, template)
difference[~valid] = 0
```

Why raised components differ. The USB Type B port, capacitors and reset button lie above the PCB plane. Viewpoint changes cause parallax and occlusion that one planar homography cannot remove. Specular reflections, shadows, exposure, residual misalignment and possible component differences also contribute. AAAA1 and ADAD1 have much larger raw MAE than AECC2–4, consistent with their stronger viewpoint and appearance changes. Bright residuals alone do not establish a manufacturing defect.

Reproducibility and references. Ten geometry tests pass, including unit-normal fitting, robust homography recovery, black-overlay preservation and valid-overlap masking. Assignment: R. Rodrigo, *EN3160 Assignment 2 on Fitting and Alignment*, 9 October 2026. Optimization background: S. Boyd and L. Vandenberghe, *Convex Optimization*, §1.1, Stanford; SciPy minimize documentation. Code, parameters and genuine commit history: `https://github.com/ahesh-dilhan/en3160-assignment-02`.